

Practical 8

AIM : K-Nearest Neighbors (K-NN)

1. Implement the K-NN algorithm for classification or regression.
2. Apply the K-NN algorithm to a given dataset and predict the class or value for test data.
3. Evaluate the accuracy or error of the predictions and analyze the results..

CODE :

```
import matplotlib.pyplot as plt

from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsRegressor
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score

# Load California Housing dataset
data = fetch_california_housing()

X = data.data
y = data.target

print("Dataset Shape:", X.shape)
print("Features:", data.feature_names)

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)

# Feature Scaling
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Create K-NN Regression model
knn_model = KNeighborsRegressor(n_neighbors=5)

# Train model
```

Name : Shubham Sanjay Karape

Roll No : T085

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```
knn_model.fit(X_train, y_train)

# Predict test data
y_pred = knn_model.predict(X_test)

# Calculate errors
mse = mean_squared_error(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

# Display Results
print("\nK-Nearest Neighbors (K-NN) Regression")
print("-----")

print("\nTraining Samples:", len(X_train))
print("Testing Samples:", len(X_test))

print("\nK Value:", 5)

print("\nMean Squared Error (MSE):", mse)
print("Mean Absolute Error (MAE):", mae)
print("R2 Score:", r2)

print("\nActual Values:")
print(y_test[:10])

print("\nPredicted Values:")
print(y_pred[:10])

# Actual vs Predicted Graph
plt.figure(figsize=(8, 5))

plt.scatter(y_test, y_pred)

plt.xlabel("Actual Values")
plt.ylabel("Predicted Values")
plt.title("K-NN Regression: Actual vs Predicted")
plt.tight_layout()
plt.show()

# Error Graph
errors = y_test - y_pred

plt.figure(figsize=(8, 5))

plt.hist(errors, bins=20)

plt.xlabel("Prediction Error")
plt.ylabel("Frequency")
plt.title("K-NN Regression Prediction Errors")
plt.tight_layout()
plt.show()
```

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OUTPUT :

```
File Edit Selection View Go Run ... AI Practicals
Explorer Problems Output Debug Console Terminal Ports
AI Practicals
  Practical3.py
  Practical4.py
  Practical5.py
  Practical6.py
  Practical7.py
  Practical8.py
  Outline
  Timeline
PS C:\Users\DELL\OneDrive\Desktop\AI Practicals> python Practical8.py
Dataset Shape: (20640, 8)
Features: ['MedInc', 'HouseAge', 'AveRooms', 'AveBedrms', 'Population', 'AveOccup', 'Latitude', 'Longitude']

K-Nearest Neighbors (K-NN) Regression
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Training Samples: 16512
Testing Samples: 4128

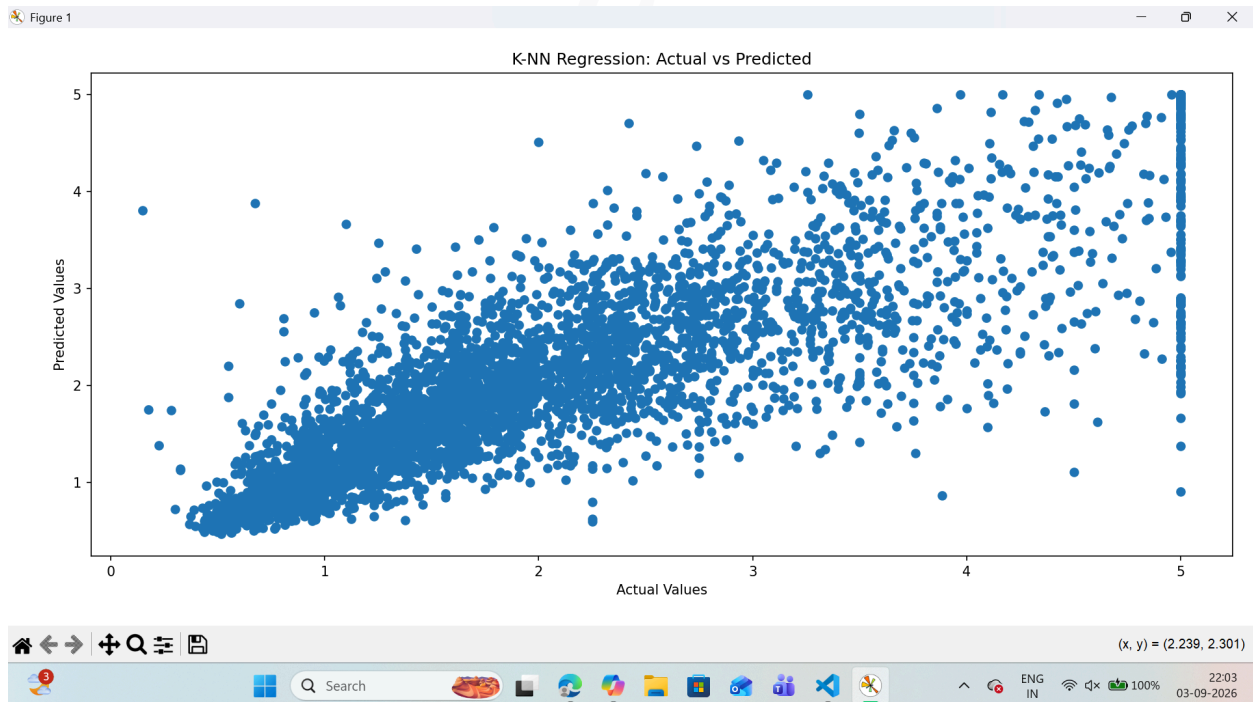
K Value: 5

Mean Squared Error (MSE): 0.4324216146043236
Mean Absolute Error (MAE): 0.4461535271317829
R2 Score: 0.6700101862970989

Actual Values:
[0.477 0.458 5.00001 2.186 2.78 1.587 1.982 1.575 3.4
 4.466 ]

Predicted Values:
[0.4988 0.7646 4.750006 2.8766 2.7262 1.7786 2.7182 1.9246
 2.0314 4.955608]
PS C:\Users\DELL\OneDrive\Desktop\AI Practicals>
```

Scatter Plot – Actual vs Predicted Values



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Histogram – Prediction Errors

